

Elad Ben Hayon

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SUMMARY

Computer Engineer with hands-on experience in embedded systems, IoT security, and hardware–software integration. Proficient in C/C++, Python, and system-level debugging/verification, with strong problem-solving skills. Experienced in designing, testing, and documenting projects involving microcontrollers, PCB design, and networking protocols.

EDUCATION

University of Massachusetts Amherst

Bachelor of Computer Engineering, GPA: 3.23

Amherst, MA

Graduated: May 2025

- **Honors:** Fall 2021 dean's list, Spring 2024 dean's list
- **Relevant Classes:** Calculus I & II, Advanced Programming, Circuits and Electronics I, Analytical Tools for ECE, Probability and Statistics, Intro to Embedded Systems, Intro to Security Engineering, Systems Programming, Computer Networking, Algorithms for ECE, Communication Systems, Intro to Quantum Computing, Signal Processing, Network Security and Privacy, Security Engineering, Operating Systems

TECHNOLOGIES

Languages: Python, C/C++, MATLAB, Verilog, SystemVerilog, Assembly (x86), Go

Tools: Docker & Docker Compose, KiCad, Quartus, VS Code, Wireshark, Quartus, Git

AI: Copilot, LLM, OpenCV

Other: Microsoft Office, Excel

Core Topics: Embedded Systems, Computer Networks, Security Engineering

PROJECTS

Bluetooth Authentication Protocol (2025)

- Developed a custom communication protocol layer for Bluetooth communication to authenticate and manage trusted devices, implemented on ESP32 microcontrollers.
- Documented protocol performance and tested resilience to unauthorized access.

Garden Protector Robot (2024-2025)

- Designed and programmed autonomous robot using an embedded system to patrol a designated area of 32 sq. feet and deter birds using motion tracking and species detection.
- Integrated sensors can detect birds along with other animals, within a 3 foot range for birds and more for larger animals, even if partially obscured, with an accuracy over 90%.
- Implemented real-time alerts with image capture via Raspberry Pi.

Low-Cost Temperature Measurement PCB (2024)

- Designed a compact (<3 sq. in.) circuit board using Kicad to measure and display temperatures between 0° to 45°C.
- Programmed in C, adhered to strict deadlines, and maintained a total cost of <15\$ per board.
- Delivered full documentation for development and assembly process.

GoFish Card Game Simulator (2023)

- Engineered game logic in C, handling edge cases and ensuring robust game play logic.

Ultrasound Distance Detector (2023)

- programmed an arduino-based system in C to measure distances up to 1.5 meters with an ultrasound sensor.
- Displayed real-time distance measurements on an OLED screen.

Projectile Motion Simulator (2022)

- Created a Matlab simulator to model projectile motion, visualizing results with graphs.

CERTIFICATIONS

Artificial Intelligence Fundamentals by IBM SkillsBuild

EXTRACURRICULAR ACTIVITIES & INTERESTS

Board Game Club: Organized tournaments and strategy workshops.

Hobbies: Science, robotics, history, reading, scuba diving (PADI/SSI certified).